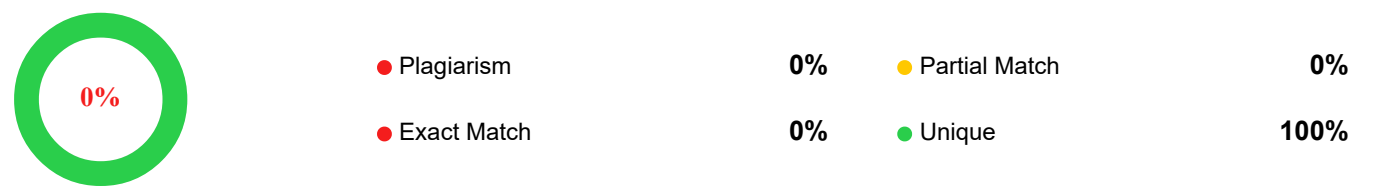




Plagiarism Detection Report by SmallSEOTOOLS



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1. PROBLEM UNDERSTANDING AND FORMULATION

1.1 Problem Definition

Sports analytics requires accurate information about the movement of athletes and sports objects such as balls. In live matches, manually observing and recording movement is time-consuming and may not provide consistent results. Therefore, a computer vision-based system is proposed to automatically track the motion of athletes or sports objects from video frames.

The proposed system processes a sports video frame by frame and identifies the movement between consecutive frames. Two computer vision approaches are considered: Optical Flow and Motion Estimation. Optical Flow provides detailed pixel-level motion information, while Motion Estimation focuses on faster identification of movement between frames.

The main engineering problem is to select an approach that provides a suitable balance between tracking accuracy and real-time processing speed. The selected method should be capable of providing useful motion information to coaches and sports analysts during live or near-live analysis.

1.2 Expected Outcomes

The expected outcomes of the proposed system are:

- Automatically process sports video frames.
- Identify the movement of athletes or sports objects.
- Estimate the direction and magnitude of motion.
- Track movement across consecutive frames.

- Visualize motion using arrows, points, bounding regions, or trajectories.
- Measure processing performance such as FPS and processing time.
- Compare Optical Flow and Motion Estimation.
- Select the most suitable approach for live sports analytics.

### 1.3 Available Information and Data

The primary input to the system is a sports video containing moving athletes or sports objects. The video consists of a sequence of image frames captured by a camera.

The available information includes:

- Video frames.
- Frame rate.
- Image resolution.
- Pixel intensity information.
- Athlete or object locations.
- Movement between consecutive frames.
- Motion direction and magnitude.
- Processing time and FPS.

The system uses these visual features to determine how objects move from one frame to another.

### 1.4 Assumptions

The following assumptions are made:

- The input video contains visible athletes or sports objects.
- Consecutive frames contain sufficient visual information for motion analysis.
- The camera provides reasonably stable video.
- The objects being tracked are visible for most of the sequence.
- Lighting conditions are reasonably suitable for computer vision processing.
- The video resolution is sufficient for detecting meaningful movement.
- The system does not require perfect identification of every individual athlete.

- Processing performance depends on the computer hardware and video resolution.

### 1.5 Constraints

The major constraints of the system are:

- Real-time performance: The system should process video quickly enough for live or near-live analysis.
- Tracking accuracy: Athlete and object movement should be tracked reliably.
- Computational cost: The selected algorithm should not require excessive processing resources.
- Occlusion: Athletes or objects may temporarily block each other.
- Camera movement: Camera panning or zooming can affect motion estimation.
- Lighting variation: Changes in brightness may influence tracking performance.
- Motion complexity: Fast athlete and object movements can make tracking difficult.
- Processing hardware: Performance is limited by available CPU/GPU resources.

### 1.6 Functional Requirements

The proposed system should perform the following functions:

1. Accept a sports video as input.
2. Extract consecutive frames from the video.
3. Preprocess the frames for motion analysis.
4. Detect or identify the athlete/object region.
5. Estimate motion using Optical Flow or Motion Estimation.
6. Track the movement across frames.
7. Generate a movement trajectory.
8. Calculate performance metrics such as FPS and processing time.
9. Display the tracking results visually.
10. Compare both approaches and identify the most suitable method.

### 1.7 Engineering Objective

The main engineering objective is to develop a computer vision-based motion tracking system that provides accurate and sufficiently fast motion information for sports analytics. The system should evaluate the trade-off between accuracy and computational speed and select an approach that satisfies the operational requirements of live sports analysis.

## 2. APPLICATION OF COMPUTER VISION KNOWLEDGE

### 2.1 Computer Vision Principles

Computer vision enables a computer to understand and analyze visual information from images and videos. In this project, a sports video is treated as a sequence of image frames. By comparing consecutive frames, the system can identify changes in the position of athletes or sports objects.

The main computer vision concepts applied are:

- Image and video representation
- Frame extraction
- Image preprocessing
- Grayscale conversion
- Feature extraction
- Motion analysis
- Object tracking
- Optical Flow
- Motion Estimation
- Trajectory analysis

### 2.2 Video Frame Processing

A video can be represented as a sequence of frames:

Video → Frame 1 → Frame 2 → Frame 3 → ... → Frame N

The system compares consecutive frames to determine how pixels or objects have moved.

For example:

Frame 1: Athlete at position A

Frame 2: Athlete at position B

The difference between these positions represents the observed motion.

### 2.3 Image Preprocessing

Before motion analysis, the input frames are preprocessed to improve computational efficiency.

The main preprocessing operations are:

1. Read the video.
2. Extract individual frames.
3. Resize frames if required.
4. Convert frames from RGB to grayscale.
5. Apply optional noise reduction.
6. Use the processed frames for motion analysis.

Grayscale images reduce the amount of data because each pixel is represented mainly by intensity instead of three colour channels.

## 2.4 Motion Information

Motion information describes how an athlete or object changes its position between consecutive frames.

Important motion properties include:

- Direction: Direction in which the object moves.
- Magnitude: Amount of movement.
- Velocity: Approximate rate of position change.
- Trajectory: Path followed by the moving object.

This information can be used to understand athlete movement and sports-object behaviour.

## 2.5 Optical Flow

Optical Flow estimates the apparent movement of pixels between two consecutive video frames.

For example:

Frame  $t \rightarrow$  Frame  $t+1 \rightarrow$  Pixel movement  $\rightarrow$  Motion vectors

Each motion vector represents the estimated direction and magnitude of movement at a particular location.

Optical Flow is useful when detailed motion information is required. It can provide dense or sparse motion information depending on the selected algorithm.

Advantages:

- Provides detailed motion information.
- Useful for analysing complex movement.

- Can capture local movement.

- Suitable for trajectory

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